1.

Exercise 5

Consider a dataset containing six one-dimensional points: $\{2, 4, 7, 8, 12, 14\}$. After three iterations of hierarchical clustering using Euclidean distance between points, we get the 3 clusters: $C_1 = \{2, 4\}$, $C_2 = \{7, 8\}$, and $C_3 = \{12, 14\}$. What is the distance between clusters C_1 and C_2 using single linkage? What is the distance between clusters C_1 and C_2 using complete linkage?

Solution:

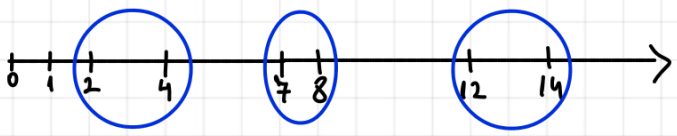
- in *single link clustering*, also called *nearest neighbor clustering*, the distance between two groups G and H is defined as the distance between the two closest members of each group:

$$d_{SL}(G, H) = \min_{i \in G, i' \in H} d_{i,i'}.$$

- in *complete link clustering*, also called *furthest neighbor clustering*, the distance between two groups is defined as the distance between the two most distant pairs:

$$d_{CL}(G, H) = \max_{i \in G, i' \in H} d_{i,i'}.$$

a) single link



$$d_{SL}(C_1, C_2) = \min_{i \in C_1, i' \in C_2} d_{i,i'} = \min\{|2-7|, |2-8|, |4-7|, |4-8|\} =$$

$$= \min\{5, 6, 3, 4\} = 3$$

b) complete link

$$d_{CL}(C_1, C_2) = \max_{i \in C_1, i' \in C_2} d_{i,i'} = \max\{|2-7|, |2-8|, |4-7|, |4-8|\} =$$

$$= \max\{5, 6, 3, 4\} = 6$$

c) average link

$$d_{AL}(C_1, C_2) = \frac{1}{m_{C_1} m_{C_2}} \sum_{i \in C_1} \sum_{i' \in C_2} d_{i,i'} = \frac{1}{2 \cdot 2}\left(|2-7| + |2-8| + |4-7| + |4-8|\right) =$$

$$= \frac{5 + 6 + 3 + 4}{4} = 4{,}5$$

La fel

Question **5**

Not yet answered

Marked out of 1.00

Flag question

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2.

10. K-means clustering algorithm

Assume we want to cluster 7 points into 3 clusters using k-means clustering algorithm. Suppose after first iteration clusters C1, C2, C3 contain the following 2-D points: C1: {(0,6),(6,0)}; C2:{(2,2),(4,4),(6,6)} and C3:{(5,5),(7,7)}

Assume we want to cluster 7 points into 3 clusters using the K-means clustering algorithm. Suppose after the first iteration, clusters C_1, C_2, and C_3 contain the following two-dimensional points:

(a) C_1 contains the 2 points: $\{(0,6),(6,0)\}$;

(b) C_2 contains the 3 points: $\{(2,2),(4,4),(6,6)\}$;

(c) C_3 contains the 2 points: $\{(5,5),(7,7)\}$

What are the cluster centers computed for these 3 clusters?

$C_1 \left(\frac{0+6}{2} ; \frac{0+6}{2} \right) \Rightarrow C_1(3,3)$

$C_2 \left(\frac{2+4+6}{3} ; \frac{12}{3} \right) \Rightarrow C_2(4,4)$

$C_3 \left(\frac{5+7}{2} ; \frac{5+7}{2} \right) \Rightarrow C_3(6,6)$ ← pt încă o iterație (6,6) ar veni la C_3 (2,2) la C_1

1. trb recalculate distanțele de la punct la noii centroizi
2. se asignează punctele
3. se calculează noii centroizi

③ ex de la curs

$\{(1,1), (1,2), (2,1), (2,2), (6,6), (6,7), (7,6), (7,7)\}$

$\mu_1^0 = (1,1)$, $\mu_2^0 = (7,7)$

$x_1 = \sqrt{(1-1)^2 + (1-1)^2} = 0$ d la μ_1

$x_2 = \sqrt{(1-1)^2 + (2-1)^2} = 1$ d la μ_1

$x_3 = \sqrt{(1-2)^2 + (1-1)^2} = 1$ d la μ_1

$x_4 = \sqrt{2}$ d la μ_1

$x_5 \longrightarrow d(x_5, \mu_1) = \sqrt{50} > d(x_5 \text{ la } \mu_2) = \sqrt{2}$

x_6
x_7
x_8 } le face la fel

μ_1^0: x_1, x_2, x_3, x_4

μ_2^0: x_5, x_6, x_7, x_8

$\mu_1^1 \left(\frac{1+1+2+2}{4} ; \frac{1+2+1+2}{4} \right) \Rightarrow \mu_1^1 (1,5 ; 1,5)$

μ_2^1 $(6,5 ; 6,5)$

se repetă calcul distanțe

$x_1 = \sqrt{(1-1,5)^2 + (1-1,5)^2} = \sqrt{0,5}$

și la celelalte

$cluster_1^1$: x_1, x_2, x_3, x_4

$cluster_2^1$: x_5, x_6, x_7, x_8

avem convergență

Inerția = $d(x_1, \mu_1^0) + d(x_2, \mu_1^0) + \ldots$

A B C b E

4.

5. Consider a dataset containing five two-dimensional points: $\{(2,10),(8,4),(5,8),(7,5),(4,9)\}$. Use average linkage to perform hierarchical clustering of the points using Euclidean distance.

Euclidean distance $d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

$d_{AB} = \sqrt{36+36} = 6\sqrt{2}$ $\quad d_{AC} = \sqrt{13}$ $\quad d_{Ab} = 5\sqrt{2}$ $\quad d_{AE} = \sqrt{5}$

$d_{BC} = 5$ $\quad d_{Bb} = \sqrt{2}$ $\quad d_{BE} = \sqrt{41}$

$d_{Cb} = \sqrt{13}$ $\quad d_{CE} = \sqrt{2}$

$d_{bE} = 5$

cea mai mică distanță este $\sqrt{2}$ $\quad d_{Bb}$ și d_{CE}

C_1: B, b

C_2: C, E

$$d(A, C_1) = \frac{1}{1 \cdot 2} \cdot (d_{AB} + d_{Ab}) = \frac{6\sqrt{2}+5\sqrt{2}}{2} = 11\frac{\sqrt{2}}{2} = 7{,}78$$

$$d(A, C_2) = \frac{1}{1 \cdot 2} (d_{AC} + d_{AE}) = \frac{\sqrt{13}+\sqrt{5}}{2} = 2{,}93$$

$$d(C_1, C_2) = \frac{1}{2 \cdot 2} (d_{CB} + d_{Cb} + d_{EB} + d_{Eb}) = 5$$

$\Rightarrow$ A e mai aproape de C_2

$\Rightarrow$ C_1: B, b

C_2: A, C, E

$$d(C_1, C_2) = \frac{1}{2 \cdot 3} (d_{AB} + d_{Ab} + d_{CB} + d_{Cd} + d_{EB} + d_{Eb}) =$$

$$= 5{,}93$$

5.

A B C B E

Use PCA to compute the first principal compo...

5. Consider a dataset containing five two-dimensional points: $\{(2,10),(8,4),(5,8),(6,4),(4,9)\}$. Use average linkage to perform hierarchical clustering of the points using Euclidean distance.

...MLP for regression, having 2 inputs, two hidden layers of 2 and 3 ...layers is ReLU. For

$d_{AB} = \sqrt{6^2+6^2} = 6\sqrt{2}$ $\quad d_{AC} = \sqrt{13}$ $\quad d_{AB} = 2\sqrt{13}$ $\quad d_{AE} = \sqrt{5}$

$d_{BC} = 5$ $\quad d_{BB} = 2$ $\quad d_{BE} = \sqrt{41}$

$d_{CB} = \sqrt{17}$ $\quad d_{CE} = \sqrt{2}$

$d_{BE} = \sqrt{29}$

cea mai mică distanță : $\sqrt{2}$ CE

$\Rightarrow C_1 : C, E$ $\quad C_1(4,5; 8,5)$

$d(A, C_1) = \frac{1}{1\cdot 2}(d_{AC} + d_{AE}) = 2,93$

$d(B, C_1) = \frac{1}{1\cdot 2}(d_{BC} + d_{BE}) = 5,7$

$d(B, C_1) = \frac{1}{1\cdot 2}(d_{BC} + d_{BE}) = 4,75$

$d_{BB} = 2$

$\Rightarrow C_2 : B, B$ $\quad C_2(7,4)$

$d(A, C_1) = 2,93$

$d(A, C_2) = \frac{1}{1\cdot 2}(d_{AB} + d_{AB}) = 7,85$

$d(C_1, C_2) = \frac{1}{2\cdot 2}(d_{BC} + d_{BE} + d_{BC} + d_{BE}) = 5,23$

$\Rightarrow$ unim A cu C_1

$C_1 : A, C, E$

$C_2 : B, B$

$d(C_1, C_2) = \frac{1}{3\cdot 2}(d_{AB} + d_{AB} + d_{BC} + d_{CB} + d_{BE} + d_{BE}) = 6,1$